Complex training

Dr. Steve Fleck, C.S.C.S.
Exercise Physiologist
U.S. Olympic Training Center
Colorado Springs, Colorado

Ken Kontor, C.S.C.S.
NSCA Executive Director

This past summer we were privi-ledged to join more than 40 fellow strength and conditioning coaches from the United States and Canada on a study tour at the Moscow Institute of Sport. The tour was sponsored by the National Strength and Conditioning Association, and focused on several relevant topics of strength training and conditioning for Soviet athletes.

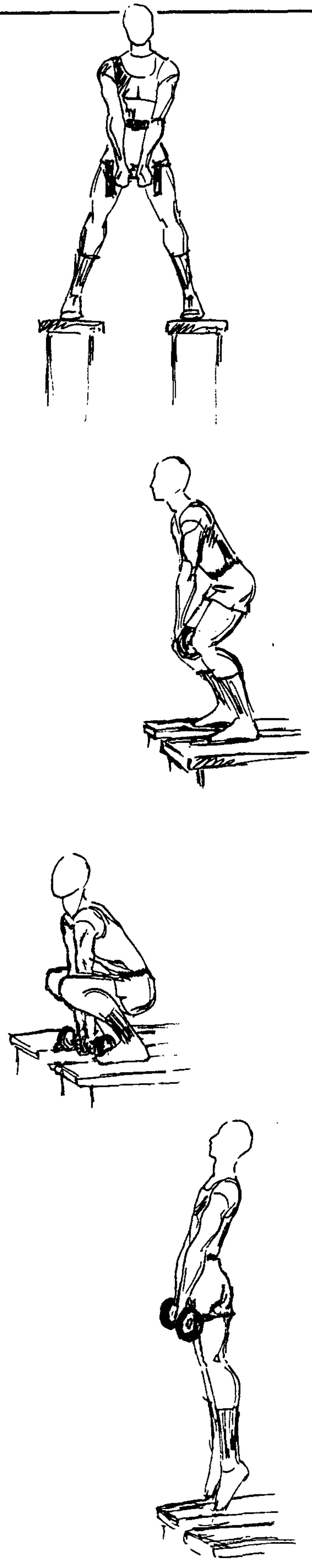
One of the interesting concepts was that of complex training. Dr. Verhoshansky defined complex training as a series of several exercises performed in succession with the goal of the entire complex the improvement of one physical characteristic. The complexes he outlined were for inclusion in the special preparatory period of training and designed to increase the ability to produce power quickly or "explosiveness" and speed of movement of the legs.

The complexes outlined all utilize muscular contractions against large resistances at relatively slow velocities of movement followed by contractions with relatively small resistances at fast velocities of movement. The contractions performed with the heavy resistances are an attempt to bring about adaptations in tension-dependent neural mechanisms that inhibit the excitation of motor neurons in voluntary maximal contractions. Neural mechanisms have been hypothesized to be in part responsible for increases in voluntary maximal contractions at slow velocities of movement in the absence of muscular hypertrophy (Costill et al, 1979; Thorstensson et al 1977, Caiozzo, Perrine & Edgerton, 1981). Precontraction of antagonistic muscles has been hypothesized to counter the inhibitory neural mechanisms in agonists (Perrine & Edgerton 1978). A strength/power training program that utilizes antagonistic precontractions is more effective in increasing low velocity strength/power than a program that does not utilize precontractions (Caiozzo et al, 1982). Thus the concept of adaptation to neural mechanisms is not unfounded.

The substantial rest periods given during the training are necessary. These complexes are designed to increase short term power output. Thus it is necessary to allow time for the replenishing of the anaerobic energy sources if the fast velocity contractions are to be performed at as high of power output as possible. Not observing the rest periods will seriously compromise the goals of the training.

Professor Verhoshansky used the example of the perception of lifting a half-full can of water when you think it's full. The excitability of the central nervous system responds in such a way that the water literally flies in the air because of the force applied. It is because if the body thinks it has to do more heavy work, so it remembers what it necessary to lift the full can and reacts accordingly.

In the following complexes, by doing a light weight after a heavy weight you fool the body into remembering the heavy weight. You, therefore, obtain high velocity of movement which will develop power.



Complex 3

Complex 1

Back squats: 2 x 2-3 repetitions at 90% of 1 RM.
Perform the concentric and eccentric portions slowly
Rest Periods: 3-4 minutes between sets
4-6 minutes after both sets
Depth Jumps: 2 x 10. Recommended height: .75 meters
Rest Periods: 3-4 minutes between sets
Perform the complex 2-3 times per training session with 8-10 minutes of rest between complexes.

Complex 2

Back Squats: 2 x 2-3 repetitions at 90% of 1 RM.
Perform the concentric and eccentric portions slowly
Rest Periods: 3-4 minutes between sets
4-6 minutes after both sets
Sequence of 5 standing long jumps: 2 x 6 repetitions (5 jumps = 1 repetition) Sequence is (L, R, L, R, both)
Rest Periods: 3-4 minutes between sets
Perform the complex 2-3 times per training session with 8-10 minutes between complexes.

Complex 3

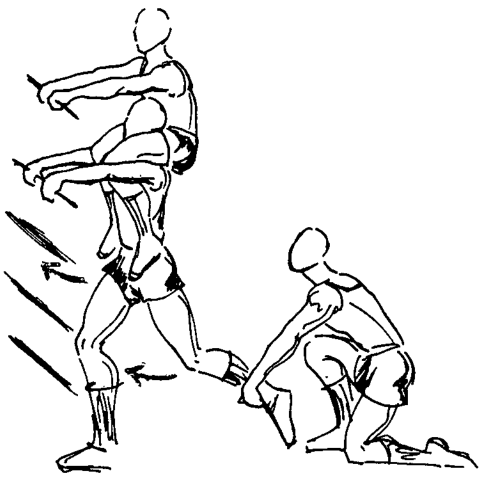
Kettle bell jumps on 2 benches: 2 x 10 jumps.
Stand on 2 benches shoulder-width apart as depicted, holding a kettle bell or dumbbell (10-35 pounds). Assume a deep squat and then jump. Perform the 10 jumps in natural rhythm.
Rest Periods: 3-4 minutes between sets
Sequence of 5 standing long jumps: 2 x 6 repetitions (perform as in complex 2)
Perform the complex 2-3 times per training session with 8-10 minutes rest between complexes

Complex 4

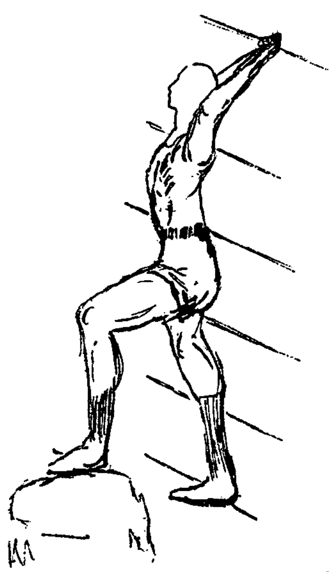
Push off and knee lift: 5-6 repetitions per leg. Position the trainee as depicted utilizing a wall rack and 2 training partners. The trainee extends his forward leg and hip and pulls his other leg through as if running. The partner on the trainee's shoulders may aid the trainee in performing leg and hip extension by pulling upwards on the wall rack. The second training partner grasps the ankle of the trainee and offers resistance to the pulling-through movement of the back leg. The trainee then returns to the original position. Perform 5-6 repetitions with one leg and then the other leg. Resistance is high at first, then reduced as leg moves through.
Rest Period: 4-6 minutes
Bounding: 40-50 meters at full speed x 3-4 repetitions
Rest Period: 3-4 minutes between repetitions
4-6 minutes after the 3-4 repetitions
Acceleration Runs: 80-100 meters, gradually increasing maximum speed for 4-5 strides then coast, x 4-5 reps
Perform the entire complex 2 times per training session during the second half of the special preparation period. Rest 8-10 minutes between complexes.

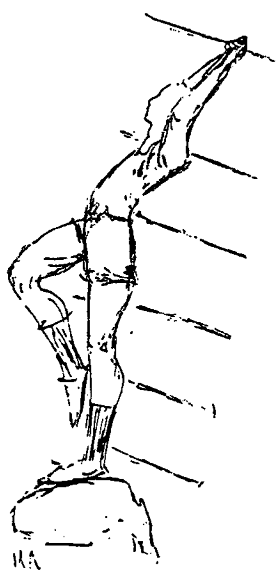
Complex 5

Leg and hip follow-through: The trainee positions himself utilizing a wall rack and gymnastics horse. He then pushes the hips forward and brings the back leg through as if running. Perform 5-6 repetitions with each leg. A weight (up to 10 pounds) or partner resistance may give resistance to the back leg for follow-through movement.
Rest Period: 4-6 minutes
30 Meter Sprints: 3-4 repetitions at maximal speed from a standing start.
Rest Period: 8-10 minutes between complexes. During rest period perform heel kicking to relax muscles.
Perform entire complex 2 times per training session.



Complex 4





Complex 5

Complex 6

Back Squats: 2 x 2-3 repetitions as in complex 1
Rest Periods: 3-4 minutes between sets
4-6 minutes after both sets
Kettle Bell Jumps: 2-3 x 8 repetitions as in complex 3, except perform them on the floor, not on 2 benches
Rest Periods: 3-4 minutes between sets
4-6 minutes after the 2-3 sets
Rest Period: 8-10 minutes between complexes
Perform the entire complex 2-3 times per training session.

Complex 7

Back Squats: 2 x 2-3 repetitions as in complex 1
Rest Periods: 3-4 minutes between sets
4-6 minutes after both sets
Back Squats: 3 x 6-8 at 30% of 1 RM, perform the repetitions at a natural tempo, the repetitions are performed explosively and the trainee should be actually jumping into the air.
Rest Periods: 3-4 minutes between sets
Perform the entire complex 2-3 times per training session resting 8-10 minutes between complexes.

Complex 8

Back Squats: 1 x 2-3 repetitions as in complex 1
Rest Period: 4-6 minutes
Back Squats: 2 x 6-8 repetitions at 30% of 1 RM as in complex 7
Rest Periods: 3-4 minutes between sets
4-6 minutes after both sets
Alternate leg bounding 5 jumps off of each leg: 2 x 5 repetitions (5 jumps off of each leg = 1 repetition)
Rest Periods: 1 minute rest between repetitions
3-4 minutes rest between sets
4-6 minutes rest after both sets
Acceleration Sprints: 3-4 x 50-60 meters
Rest Periods: between sprints 10-15 seconds

Perform the entire complex 2 times per training session, resting 6-8 minutes between complexes. After this complex some non-intensive activity such as (non-emotional) basketball for 5-10 minutes is recommended.

The complexes outlined can be divided into three types: 1) heavy loaded contractions followed by contractions against body weight, 2) heavy loaded contractions followed by contractions against lighter loading, 3) a combination of 1 and 2. Complex training should be performed 2 to 3 times per week during the special preparatory period of training.

Complexes can be developed for different muscle groups or for specific sports. Creativity and imagination are the keys. ●

References

1. Caiozzo, V.J., Laird, T., Chow, K., Prietto, C.A., and McMaster, W.C. 1982. The use of precontractions to enhance the in-vivo force-velocity relationship. ***Medicine and Science in Sports and Exercise*** 14:162.
2. Caiozzo, V.J., Perrine, J.J., and Edgerton, V.R. 1981. Training induced alterations of the in-vivo force-velocity relationship of human muscle. ***Journal of Applied Physiology: Respiratory, Environmental and Exercise Physiology*** 51:750-754.
3. Costill, D.L., Coyle, E.F., Fink, W.F., Lesmes, G.R., and Witzman, F.A. 1979. Adaptation in skeletal muscle following strength training. ***Journal of Applied Physiology: Respiratory, Environmental and Exercise Physiology*** 46:96-99.
4. Perrine, J.J., and Edgerton, V.R. 1978. Muscle force-velocity and power-velocity relationships under isokinetic loading. ***Medicine and Science in Sports*** 10:159-166.